(L)300mm

(W)200mm

pharmaniaga®

Clarithromycin

Tablet

DESCRIPTION

Pharmaniaga Clarithromycin Tablet 250 mg
Yellow, oblong, convex, film-coated with 'RMB' marking on one side and breakline on the other.

Pharmaniaga Clarithromycin Tablet 500 mg
Yellow, oblong, convex, film-coated, plain on one side and Pharmaniaga icon on the other.

COMPOSITION

Pharmaniaga Clarithromycin Tablet 250 mg AND 500 mg
Each tablet f/c contains Clarithromycin 250 mg and 500 mg.

PHARMACODYNAMICS

Clarithromycin is an antibiotic of the macrolide family. Clarithromycin exerts its antibacterial activity by inhibiting the synthesis of proteins, by means of a link with the 50S sub-unit of the cellular ribosome. Clarithromycin has demonstrated excellent *in-vitro* activity against both standard strains of bacteria and clinical isolates. It is highly potent against a wide variety of aerobic and anaerobic gram-positive and gram-negative organisms. The *in-vitro* antibacterial spectrum of clarithromycin is as follows:

Usually Sensitive Bacteria: *Streptococcus agalactiae, Streptococcus pyrogenes, Streptococcus viridans, Streptococcus pneumoniae, Haemophilus influenzae, Haemophilus parainfluenzae, Neisseria gonorrheae, Listeria monocytogenes, Legionella pneumophila, Pasteurella multocida, Mycoplasma pneumoniae, Helicobacter (Campylobacter) pylori, Campylobacter jejuni, Chlamydia trachomatis, Chlamydia pneumoniae (TWAR), Moraxella (Branhaemella) catarrhalis, Bordetella pertussis, Borrelia burgdorferi, Staphylococcus aureus, Clostridium perfringens, Peptococcus niger, Propionibacterium acnes, Bacteroides melaninogenicus, Mycobacterium avium, Mycobacterium leprae, Mycobacterium kansasii, Mycobacterium chelonae, Mycobacterium fortuitum, Mycobacterium intracellulare.*

Non-Sensitive Bacteria: Enterobacteriaceae, *Pseudomonas* sp.

PHARMACOKINETICS

The principal metabolite of clarithromycin in man and other primates is a microbiologically-active metabolite, 14-OH-clarithromycin. This metabolite is as active or 1- to 2- fold less active than parent compound for most organisms, except for *H. influenzae* against which it is twice as active. The parent compound and the 14-OH-metabolite exert either an additive or synergistic effect on *H. influenzae in vitro* and *in vivo*, depending on bacterial strains.

Clarithromycin is rapidly absorbed from the gastrointestinal tract after oral administration. The absolute bioavailability of 250 mg clarithromycin was approximately 50%. Food slightly delays both the onset of Clarithromycin absorption and the formation of the antimicrobially active metabolite, 14-OH clarithromycin, but does not affect the extend of bioavailability. Therefore, Clarithromycin tablets may be given without regard to food.

In fasting healthy human subjects, peak serum concentrations were attained within 2 hours after oral dosing. Steady state peak serum clarithromycin concentrations were attained in 2 to 3 days and were approximately 1 µg/mL with a 250 mg dose administered every 12 hours, 2 to 3 µg/mL with a 500 mg dose administered every 12 hours and 3 to 4 µg/mL with a 500 mg dose administered every 8 hours. The elimination half-life of clarithromycin was about 3 to 4 hours with 250 mg administered every 12 hours but increased to 5 to 7 hours with 500 mg administered every 8 to 12 hours.

The nonlinearity of clarithromycin pharmacokinetics is slight at the recommended dose of 250 mg and 500 mg administered every 8 to 12 hours. With a 250 mg every 12 hours dosing, the principal metabolite, 14-OH clarithromycin, attains a peak steady-state concentration of about 0.6 µg/mL and has an elimination half life of 5 to 6 hours. With a 500 mg every 8 to 12 hours dosing, the peak steady-state concentration of 14-OH clarithromycin is slightly higher (up to 1 µg/mL), and its elimination half-life is about 7 to 9 hours. With any of these dosing regimens, the steady-state concentration of the metabolite is generally attained within 2 to 3 days.

After a 250 mg tablets every 12 hours, approximately 20% of the dose is excreted in the urine as clarithromycin while after a 500 mg tablets every 12 hours, the urinary excretion of clarithromycin is somewhat greater, approximately 30%.

The renal clearance of clarithromycin is, however, relatively independent of the dose size and approximates the normal glomerular filtration rate. The major metabolite found in urine is 14-OH clarithromycin, which accounts for an additional 10% to 15% of the dose with either a 250 mg or a 500 mg tablet administrated every 12 hours.

Steady-state concentrations of clarithromycin and 14-OH clarithromycin observed following administration of 500 mg doses of clarithromycin every 12 hours to adult patients with HIV infection were similar to those observed in healthy volunteers. In adult HIV-infected patients taking 500 or 1000 mg doses of clarithromycin every 12 hours, steady-state clarithromycin Cmax values ranged from 2 to 4 µg/mL and 5 to 10 µg/mL, respectively.

The steady-state concentrations of clarithromycin in subjects with impaired hepatic function did not differ from those in normal subjects; however, the 14-OH clarithromycin concentrations were lower in the hepatically impaired subjects. The decreased formation of 14-OH clarithromycin was at least partially offset by an increase in renal clearance of clarithromycin in the subjects with impaired hepatic function when compared to healthy subjects.

The pharmacokinetics of clarithromycin was also altered in subjects with impaired renal function. Clarithromycin and the 14-OH clarithromycin metabolite distribute readily into body tissues and fluids. There are no data available on cerebrospinal fluid penetration. Because of high intracellular concentrations, tissues concentrations are higher than serum concentrations.

INDICATIONS

Treatment of infection caused by pathogens sensitive to Clarithromycin. Infections of nasopharynx tract (tonsillitis, pharyngitis) and of paranasal sinuses. Infections of lower respiratory tract: bronchitis, bacterial pneumonia, and atypical pneumonia. Skin infections: impetigo, erysipelas, folliculitis, furunculosis and septic wounds.

CONTRAINDICATIONS

Clarithromycin is contraindicated in patients with a known hypersensitivity to clarithromycin, erythromycin, or any of the macrolide antibiotics.

Concomitant administration of clarithromycin with cisapride, pimozide, or terfenadine is contraindicated. Clarithromycin and ergot derivatives should not be co-administered.

CLARITHROMYCIN SHOULD NOT BE USED IN PREGNANT WOMEN EXCEPT IN CLINICAL CIRCUMSTANCES WHERE NO ALTERNATIVE THERAPY IS APPROPRIATE. IF PREGNANCY OCCURS WHILE TAKING THIS DRUG, THE PATIENT SHOULD BE APPRISED OF THE POTENTIAL HAZARD TO THE FETUS. CLARITHROMYCIN HAS DEMONSTRATED ADVERSE EFFECTS OF PREGNANCY OUTCOME AND/OR EMBRYO-FETAL DEVELOPMENT IN MONKEYS, RATS, MICE, AND RABBITS AT DOSES THAT PRODUCED PLASMA LEVELS 2 TO 17 TIMES THE SERUM LEVELS ACHIEVED IN HUMANS TREATED AT THE MAXIMUM RECOMMENDED HUMAN DOSES.

There have been reports of an interaction between erythromycin and astemizole resulting in QT prolongation and torsade de pointes. Concomitant administration of erythromycin and astemizole is contraindicated. Because clarithromycin is also metabolised by cytochrome P450, concomitant administration of clarithromycin with astemizole is not recommended.

Concomitant administration of clarithromycin and the following drugs is contraindicated: Domperidone as this may result in QT prolongation and cardiac arrhythmias including ventricular tachycardia, ventricular fibrillation, and torsades de pointed (See Section INTERACTIONS WITH OTHER MEDICAMENTS)

ADVERSE REACTIONS

After oral administration of clarithromycin, some cases of G.I disturbances have been reported (nausea, vomiting, heartburns, abdominal pain, diarrhoea), headache and skin rashes. Other adverse reactions reported include stomatitis, glossitis, oral monilia, taste perversion pseudomembranous colitis.

Transient CNS events including anxiety, behavioural changes, confusional states, depersonalisation, disorientation, hallucinations, insomnia, manic behaviour, nightmares, psychosis, tinnitus, tremor and vertigo have been reported during post-marketing surveillance. Events usually resolve with discontinuation of the drug.

Hepatic dysfunction, including increased liver enzymes, and hepatocellular and/or cholestatic hepatitis, with or without jaundice, has been infrequently reported with clarithromycin. This hepatic dysfunction may be severe and is usually reversible. In very rare instances, hepatic failure with fatal outcome has been reported and generally has been associated with serious underlying diseases and/or concomitant medications.

There have been rare reports of hypoglycaemia, some of which have occurred in patients taking oral hypoglycaemic agents or insulin. Rarely, erythromycin and clarithromycin have been associated with ventricular arrhythmias, including ventricular tachycardia and torsade de pointes, in individuals with prolonged QT_c intervals.

Skin and subcutaneous Tissue Disorders
Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Immunocompromised Paediatric Patients
In AIDS and other immunocompromised patients treated with the higher doses of clarithromycin over long periods of time for mycobacterial infections, it is often difficult to distinguish adverse events possibly associated with clarithromycin administration from underlying signs of HIV disease or intercurrent illness.

The most frequently reported adverse events excluding those due to the patient's concurrent condition, were tinnitus, deafness, vomiting, nausea, abdominal pain, purpuric rash, pancreatitis, and increased amylase.

Evaluations of laboratory values for these patients were made by analysing those values outside the seriously abnormal level (i.e. the extreme high or low limit) for the specified test. Based on these criteria, one paediatric AIDS patient receiving < 15 mg/kg/day of clarithromycin had seriously abnormal (elevated) total bilirubin; of the patients receiving 15 to < 25 mg/kg/day of clarithromycin, there was one each

reported as seriously abnormal SGPT, BUN, and seriously decreased platelet count. None of these seriously abnormal values for these laboratory parameters were reported for patients received the highest dosage, (≥ 25 mg/kg/day) of clarithromycin.

attn	customer **Pharmaniaga Manufacturing Berhad**	date **27.11.2020**
1 colours	process colours: Black	**Please Chop & Sign For Approval**
size description	(L)**300 x** (W)**200mm** **Clarithromycin Leaflet – Front**	
FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my	
artwork prepared by: cynthia yap	email: graphic@focusprint.info (graphic Dept)	**date:**

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ARTWORK LOG

Revision no.	Date	Reason for Change
01	23.04.14	- Amend text (25°C to 30°C)at back page & PRP no
02	15.05.14	- Amend Storage Conditions & PRP no.
03	16.05.14	- Amend Storage Conditions.
04	22.09.14	- Amend text in description.
05	01.10.14	- Amend text in Indication & Adverse Reactions.
06	03.04.15	- AAdd in Date of Revision & amend Description.
07	30.01.18	- Amend text.
08	08.02.18	- Amend text.
09	31.12.18	- New artwork received.
10	03.01.19	- Amend text.
11	03.01.19	- Amend text.
12	08.05.2020	- Amend text and revision date.
13	27.11.2020	- Amend text and revision date.

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Effects on ability to drive and operate machines
There are no data on the effect of clarithromycin on the ability to drive or use machines. The potential for dizziness, vertigo, confusion and disorientation, which may occur with the medication, should be taken into account before patients drive or use machines.

PRECAUTIONS
The physician should not prescribe clarithromycin to pregnant women without carefully weighing the benefits against risk, particularly during the first three months of pregnancy.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including clarithromycin, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents.

Clarithromycin is principally excreted by the liver. Therefore, caution should be exercised in administering the antibiotic to patients with impaired hepatic function. Caution should also be exercised when administering clarithromycin to patients with moderate to severe renal failure. Attention should also be paid to the possibility of cross-resistance between clarithromycin and other macrolide drugs, as well as lincomycin and clindamycin.

Clarithromycin in combination with ranitidine bismuth citrate therapy is not recommended in patients with creatinine clearance less than 25 mL/min.

Clarithromycin in combination with ranitidine bismuth citrate should not be used in patients with a history of acute porphyria.

In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Steven-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP)], clarithromycin should be discontinued immediately and appropriate treatment should be urgently initiated.

Pregnancy
Refer to contraindications

Lactation
It is not known whether clarithromycin is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when clarithromycin is administered to a nursing woman.

Paediatric
Safety and effectiveness of clarithromycin in paediatric patients under 6 months of age have not been established.

Geriatric
Kidney function declines progressively with age. This condition can affect pharmacokinetics profile. Therefore, dosage adjustment should be considered in elderly patients with several renal impairment.

INTERACTIONS WITH OTHER MEDICAMENTS
Theophylline
Clarithromycin use in patients who are receiving theophylline may be associated with an increase of serum theophylline concentrations. Monitoring of serum theophylline concentrations should be considered for patients receiving high doses of theophylline or with baseline concentrations in the upper therapeutic range.

Carbamazepine
Concomitant administration of single doses of clarithromycin and carbamazepine has been shown to result in increased plasma concentrations of carbamazepine. Blood level monitoring of carbamazepine may be considered.

Terfenadine
When clarithromycin and terfenadine were co-administered, plasma concentrations of the active acid metabolite of terfenadine were threefold higher, on average, than the values observed when terfenadine was administered alone. The pharmacokinetics of clarithromycin and the 14-hydroxy- clarithromycin were not significantly affected by co-administration of terfenadine once clarithromycin reached steady-state conditions. Concomitant administration of clarithromycin with terfenadine is contraindicated.

Omeprazole
Clarithromycin 500 mg every 8 hours was given in combination with omeprazole 40 mg daily to healthy adult subjects. The steady-state plasma concentrations of omeprazole were increased by the concomitant administration of clarithromycin.

Zidovudine
Simultaneous oral administration of clarithromycin tablets and zidovudine to HIV-infected adult patients resulted in decreased steady-state zidovudine concentrations.

Oral anticoagulants
Concomitant administration of clarithromycin and oral anticoagulants may potentiate the effects of the oral anticoagulants. Prothrombin times should be carefully monitored while patients are receiving clarithromycin and oral anticoagulants simultaneously.

Elevated digoxin serum concentrations in patients receiving clarithromycin and digoxin concomitantly have also been reported. Some patients have shown clinical signs consistent with digoxin toxicity, including potentially fatal arrhythmias. Serum digoxin levels should be carefully monitored while patients are receiving digoxin and clarithromycin simultaneously.

The following drugs or drug classes are known or suspected to be metabolised by the same CYP3A isoxyme: alprazolam, astemizole, carbamazepine, cisapride, cyclosporine, disopyramide, ergot, alkaloids, lovastatin, methylprednisolone, midazolam, omeprazole, oral anticoagulants (e.g. warfarin), pimozide, quinidine, rifabutin simvastatin, tacrolimus, terfenadine and triazolam.

Co-administration of clarithromycin, known to inhibit CYP3A, and a drug primarily metabolized by CYP3A may be associated with elevations in drug concentrations that could increase or prolong both therapeutic and adverse effects of the concomitant drug.

The following drugs or drug classes are known or suspected to be metabolized by CYP3A isozyme: Domperidone.

Use of clarithromycin in patients concurrently taking drugs metabolized by the cytochrome P450 system (e.g. Warfarin, ergot alkaloids, midazolam, lovastatin, phenytoin ciclosporin) may be associated with an increase in the serum levels of these drugs.

Quinidine/ disopyramide have been reported to cause torsade de pointes when taken in combination with clarithromycin.

DOSAGE AND ADMINISTRATION
Clarithromycin recommended dosage in adults is one 250mg tablet every 12 hours. In cases of severe infections, dosage can be increased up to 500 mg every 12 hours.

Administration must be continued, according to severity of infection, up to 6-14 days.

In patients with renal impairment with creatinine clearance less than 30 mL/min, the dosage should be reduced by one-half. (Dosage should not be continued beyond 14 days in these patients).

OVERDOSAGE
Reports indicate that the ingestion of large amounts of clarithromycin can be expected to produce gastrointestinal symptoms. One patient who had a history of bipolar disorder ingested 8 g of clarithromycin and showed altered mental status, paranoid behaviour, hypokalaemia, and hypoxemia. Allergic reactions accompanying overdosage should be treated by the prompt elimination of unabsorbed drug and supportive measures.

As with other macrolides, clarithromycin plasma levels are not expected to be appreciably affected by haemodialysis or peritoneal dialysis.

ROUTE OF ADMINISTRATION
Oral

STORAGE CONDITIONS
Store below 30°. Protect from light.

SHELF LIFE
Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION
Pharmaniaga Clarithromycin Tablet 250 mg
In blisters of 10, 14, 28, 30, 100 and 140 tablets.

Pharmaniaga Clarithromycin Tablet 500 mg
In blisters of 10, 14, 28, 100 and 140 tablets.

Date of revision: 26th November 2020

PRODUCT REGISTRATION HOLDER/
MANUFACTURER:
PHARMANIAGA MANUFACTURING BERHAD
(198001006232)
11A, Jalan P/1, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia.

PRP 0407.9/ BRP 0407.4/ PRP 0408.9/ PRP 0409.8/ PRP 0410.8
261120

attn	customer **Pharmaniaga Manufacturing Berhad**	date **27.11.2020**
1 colours	process colours: Black	**Please Chop & Sign For Approval**
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FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my	
artwork prepared by: cynthia yap	email: graphic@focusprint.info (graphic Dept)	date:

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